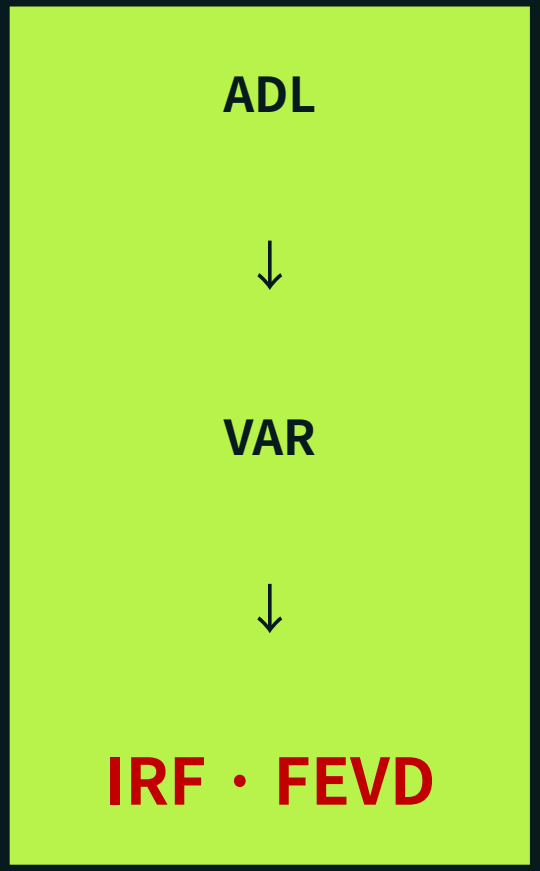
The VAR Model
Interacting Time Series

A system of autoregressions — one equation per variable



Four steps cover the structure and uses of the VAR

01	Basics	**Why a VAR** Sims (1980) · endogeneity · reduced form
02	Model and estimation	**An ADL for every variable** Bivariate VAR(1) · matrix form · parameter count
03	Stability and analysis	**When shocks die out** Eigenvalues · Granger · IRF · FEVD
04	Cases and lab	**From monetary policy to KOSPI** Price puzzle · return news · Claude Code

Concept → model → stability → analysis tools → cases

A VAR does not decide in advance which variables are exogenous

Sims (1980) — macro variables influence one another

PROBLEM

No clear exogeneity

- Rates, prices, output interact
- Structural models rely on ad hoc restrictions

IDEA

Let the data speak

- Minimal theory restrictions
- Treat all variables symmetrically

PAYOFF

One model, three uses

- Forecasting
- Granger causality
- Impulse response analysis

VAR = a system that treats every variable as endogenous and explains each by the lags of all

A reduced-form VAR builds each equation from lagged values only

This deck covers the reduced form; the structural VAR comes in the next deck

ENDOGENOUS

Endogenous variable

- Determined within the model
- In a VAR: every variable

REDUCED FORM

Reduced-form VAR

- Only lags on the right side
- OLS equation by equation

STRUCTURAL · SVAR

Structural VAR

- Identifies same-period effects
- Needs extra restrictions

The reduced form alone cannot tell the direction of same-period causation

Plotting IRFs therefore requires orthogonalization or identification

A VAR stacks one ADL equation for each variable

ADL (autoregressive distributed lag) — explains a variable by its own lags and another's lags

ADL — one equation

Explains y; x comes from outside

$$y_t \leftarrow y_{t-1},\ x_{t-1}$$

x_t = exogenous, given

One equation

per variable

VAR — N equations, all endogenous

y equation

$$y_t \leftarrow y_{t-1},\ x_{t-1}$$

x equation

$$x_t \leftarrow x_{t-1},\ y_{t-1}$$

Same regressors: all variables' lags

The two equations of a Granger causality test are exactly a bivariate VAR

A bivariate VAR(1) regresses each variable on both lagged values

Two variables y and x, one lag — the simplest VAR

Bivariate VAR(1) system

$$y_t = a + b_1\, y_{t-1} + c_1\, x_{t-1} + e_t$$
$$x_t = a^* + b_1^*\, x_{t-1} + c_1^*\, y_{t-1} + e_t^*$$

Own-lag coefficients

Effect of own past

Cross-lag coefficients

Nonzero → Granger

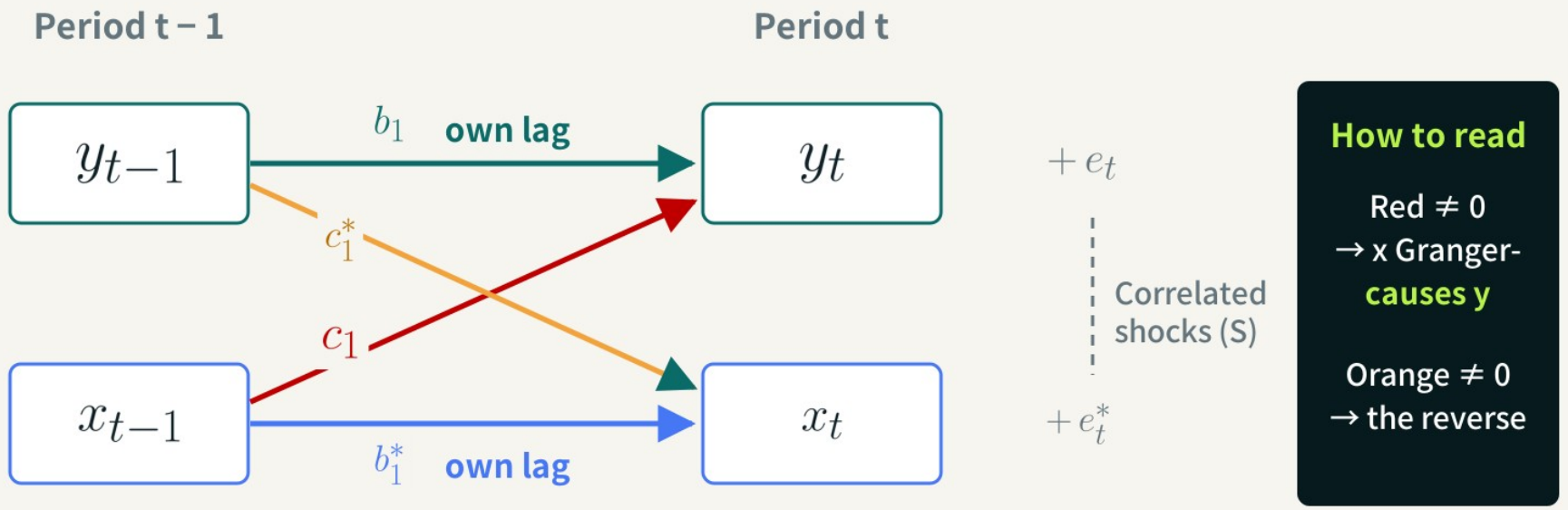
Equation shocks

May be correlated

Even with a nonzero shock covariance, the reduced form cannot say which side is the cause

The four coefficients are four arrows from last period to this period

Straight arrows are own-lag coefficients; crossing arrows are cross-lag coefficients



The two crossing arrows carry the dynamic interaction between the variables

In matrix form, a VAR with any number of lags has the same shape

The vector z is (y, x)′ — row 1 of the coefficient matrix is the y equation, row 2 the x equation

Matrix form — VAR(1)

$$z_t = \begin{pmatrix} y_t \\ x_t \end{pmatrix} = \begin{pmatrix} a \\ a^* \end{pmatrix} + \underbrace{\begin{pmatrix} b_1 & c_1 \\ c_1^* & b_1^* \end{pmatrix}}_{A_1} z_{t-1} + \begin{pmatrix} e_t \\ e_t^* \end{pmatrix}$$

VAR(p) — p lags, intercept k, shock covariance S

$$z_t = k + A_1 z_{t-1} + A_2 z_{t-2} + \cdots + A_p z_{t-p} + e_t, \qquad \mathrm{Cov}(e_t) = S$$

The diagonal of S holds shock variances; the off-diagonal, same-period covariances

The number of coefficients grows quickly with variables and lags

Each equation has one intercept and Np coefficients — N variables, p lags

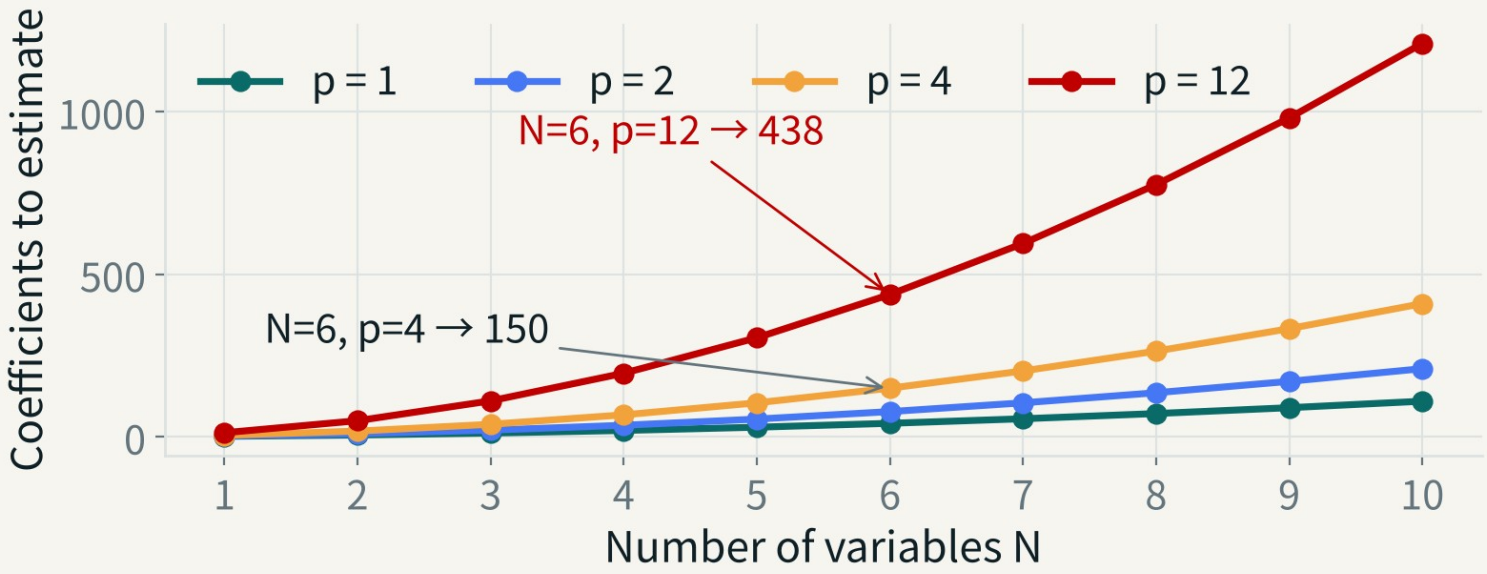
Total coefficients

$$N\,(1 + Np)$$

Monthly data example

N=6, p=4 → **150**

N=6, p=12 → **438**



Curse of dimensionality — in small samples keep the 4–6 variables the question needs

Equation-by-equation OLS is efficient for a VAR

All equations share the same regressors, so OLS equals SUR (GLS)

ESTIMATION

OLS per equation

- Identical regressor set
- OLS = SUR, efficient

LAG SELECTION

Lag length

- Compare AIC · BIC · HQ
- BIC usually picks fewer lags

NONSTATIONARY

If variables are I(1)

- VAR in differences
- VECM if cointegrated

Asset-pricing VARs use stationary variables such as returns and spreads

Shocks die out when all eigenvalues lie inside the unit circle

Stability — eigenvalues of the coefficient matrix (VAR(1)) or of the companion matrix (VAR(p))

Stability condition — every eigenvalue below 1 in modulus

$$\text{all } |\lambda_i(\text{companion})| < 1$$

Worked example (variable order y, x)

$$A_1 = \begin{pmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{pmatrix} \implies \lambda = 0.6,\ 0.3$$

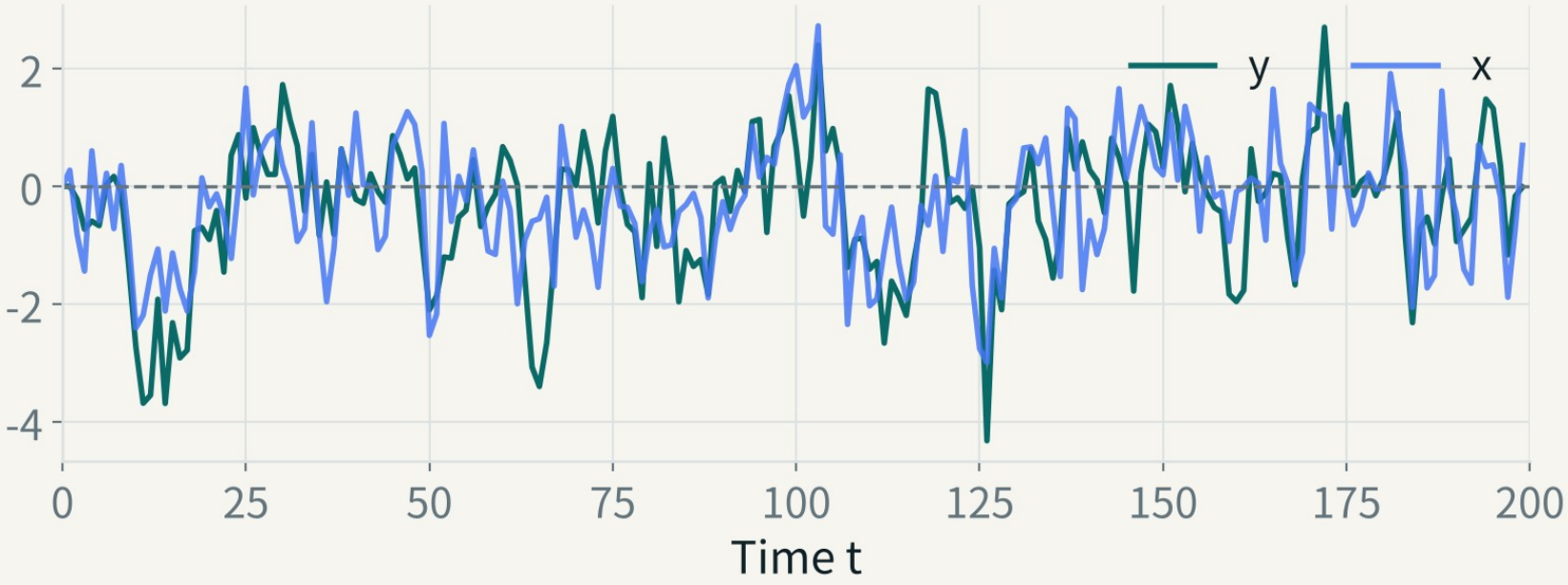
How to read

- Both eigenvalues < 1 → stable
- The larger root 0.6 sets the decay speed
- Half-life ≈ 1.4 periods

Impulse responses and forecasts are meaningful only for a stable VAR

A stable VAR(1) moves around its mean with the two series in step

200 periods generated from the worked-example coefficient matrix — shock correlation 0.4



Through each other's lags and correlated shocks, the two series swing in the same direction

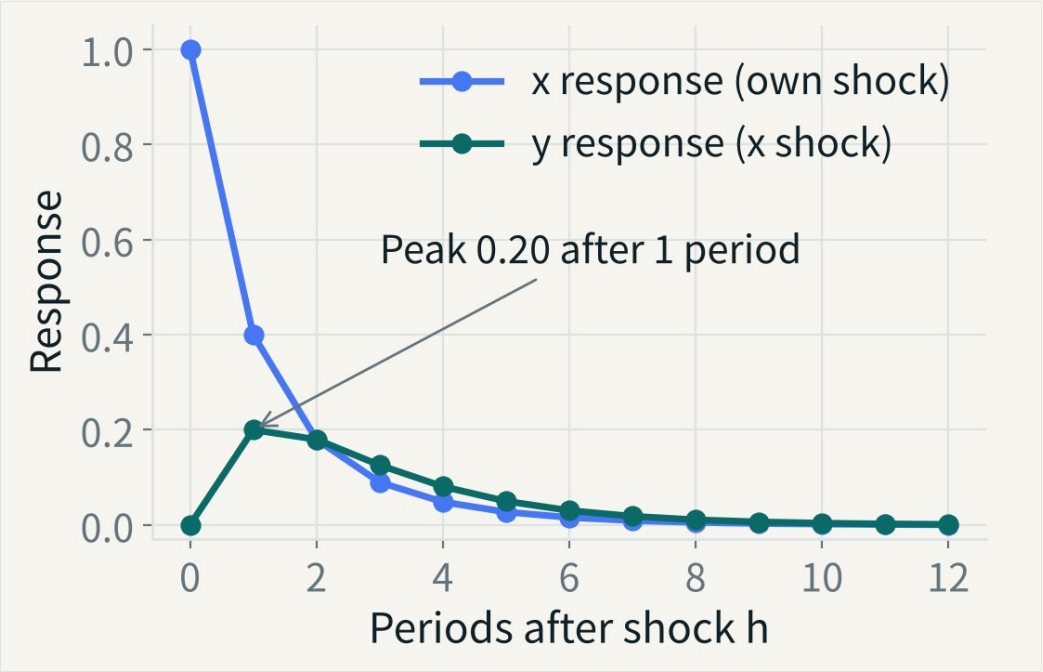
※ Simulated data — coefficients 0.5, 0.2, 0.1, 0.4; shock variance 1, correlation 0.4; seed 7

A unit shock to x reaches y most strongly one period later

Non-orthogonalized response — shock correlation ignored, no further shocks

Results only

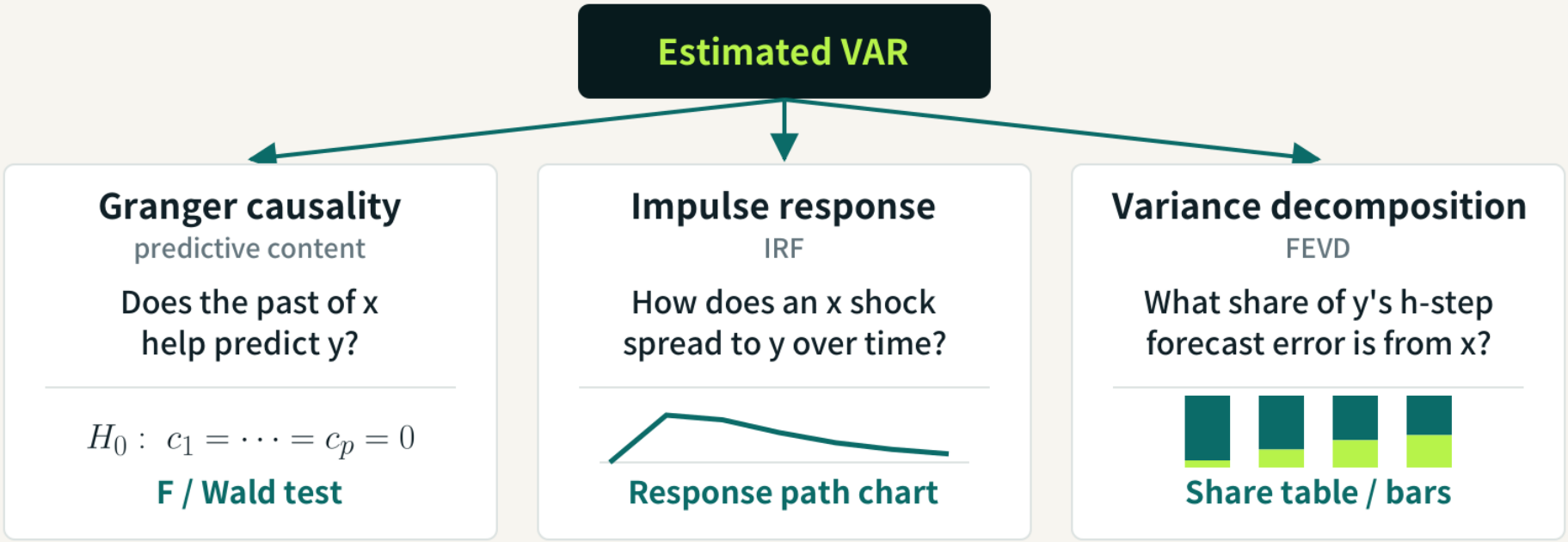
- y, h=1: 0.20
- y, h=2: 0.18
- y, h=3: 0.126
- x: 0.4 → 0.18



The hand calculation is in the workbook deck — here we read only the shape of the path

One estimated VAR delivers Granger causality, IRFs and FEVD

Three questions asked in turn of the same reduced-form VAR



Report IRF and FEVD charts and tables, not coefficient tables

Cholesky orthogonalization results depend on the variable ordering

Reduced-form shocks are correlated, so a 'pure x shock' must be defined separately

CHOLESKY DECOMPOSITION

Earlier → later

- Earlier ones act first
- Reorder → new IRF

Order = exogeneity assumption

ALTERNATIVES

Less order-dependence

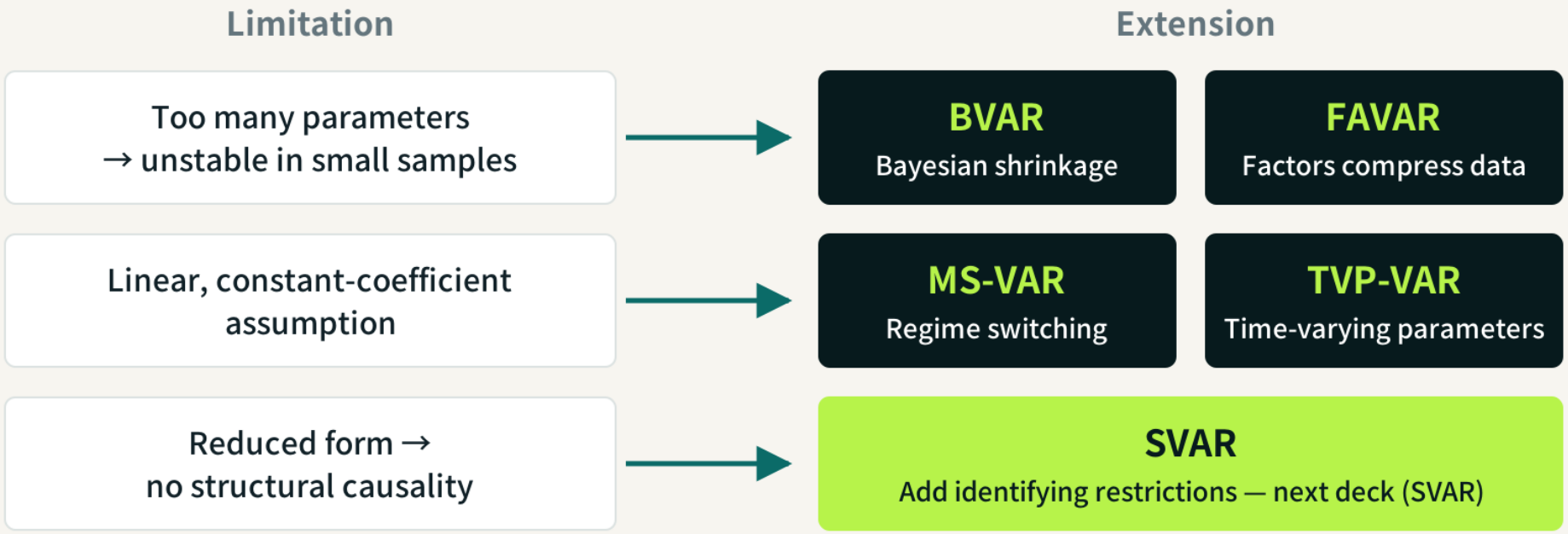
- Generalized IRF
- SVAR sign restrictions

Identify from theory

Choose the ordering by economic reasoning, not by statistics

Each of three VAR limitations has led to an extension

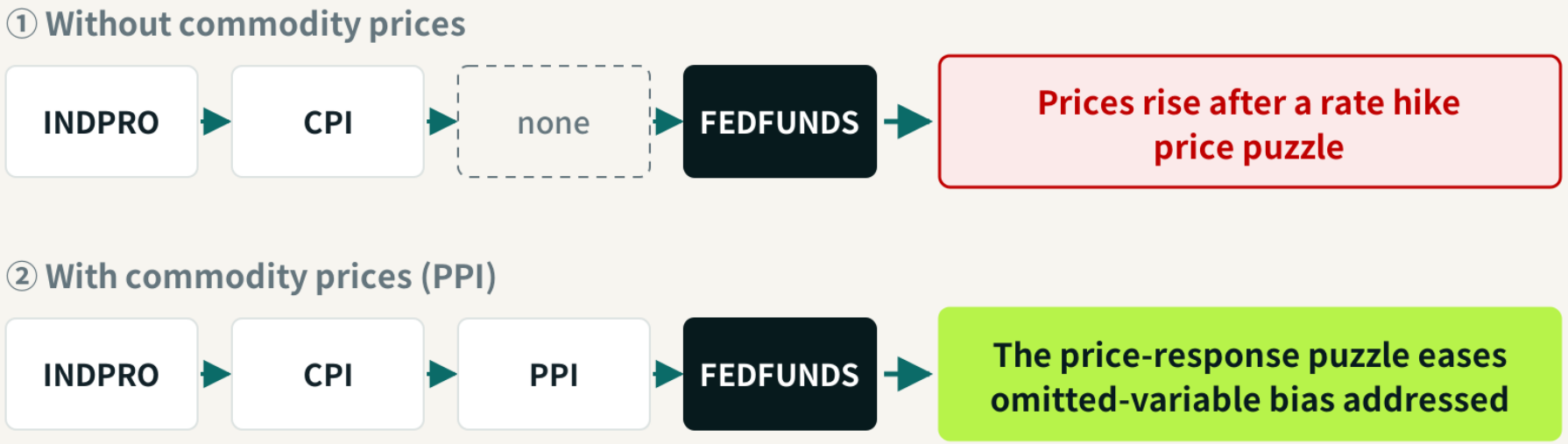
BVAR · FAVAR · MS-VAR · TVP-VAR · SVAR



Reduced-form Granger causality is predictive power, not structural causation

Adding commodity prices eases the price puzzle in a monetary policy VAR

Case ① output, prices, (commodity prices), policy rate — Christiano–Eichenbaum–Evans (1999), Sims (1992)



Box order = Cholesky order — earlier variables move later ones contemporaneously

After a tightening shock, output declines over several quarters

A single omitted variable can flip the sign of an IRF

A VAR splits stock-price moves into two kinds of news

Case ② VAR of returns, dividend yield and term spread — Campbell (1991)

Unexpected return = cash-flow news − discount-rate news

$$r_{t+1} - E_t r_{t+1} = N_{CF,t+1} - N_{DR,t+1}$$

UNEXPECTED RETURN

Unexpected return

Miss vs VAR forecast

CASH-FLOW NEWS

Cash-flow news

Dividend revisions

DISCOUNT-RATE NEWS

Discount-rate news

The larger share

The VAR forecasting equations are the engine that computes revisions in expectations

Return predictability lowers equity risk for long-horizon investors

Case ③ short rate, stock and bond returns, dividend yield, term spread — Campbell & Viceira (2002)

ONE-PERIOD VIEW

Short horizon

- Risk = 1-period vol
- Scales with horizon

Assumes i.i.d. returns

VAR VIEW · SAA

Long horizon

- Predictable returns
- Risk by horizon

Relative equity risk falls with horizon

Long-horizon investors such as pension funds estimate the term structure of risk with a VAR

Higher US rates pass through won depreciation to a falling KOSPI

Case ④ VAR of US 10Y yield, KRW/USD and KOSPI — macro-financial links in Korea



Transmission path traced by the IRF

Cholesky order = exogeneity order

US rate → KRW/USD → KOSPI: Korean variables cannot move the US rate within the same period

IRF results are credible only when variables are ordered by exogeneity

※ Practice check: 4–6 variables the question needs, stationary series, report with IRF and FEVD charts

Two prompts hand the whole VAR analysis to Claude Code

Students paste the prompts and check results — Claude Code downloads, runs and fixes errors

① US monetary policy VAR and the price puzzle — FRED monthly, 1965–2007

Pull 4 FRED monthly series → Logs; rate in levels → Fit VAR BIC, stability → Rate-shock IRF 48 months → Add PPI md report

② US rate → KRW/USD → KOSPI spillover — FRED and yfinance weekly, 2010–

US 10Y, FX, KOSPI weekly → bp changes, log returns → Ordered VAR cumulative IRF → Re-estimate post-2022 → 3-line takeaway for the IC

The key line ends each prompt: 'Fix any errors yourself and tell me what you changed'

Treat every variable as endogenous.

Explain each by the lags of all.

Read results through IRF and FEVD.

Next deck — VAR and SVAR:
identifying same-period causation